
EFFECT OF MICROPLASTICS ON WATER QUALITY AND *OREOCHROMIS NILOTICUS* (NILE TILAPIA) IN JAKARA RIVER, KANO STATE, NIGERIA

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ABSTRACT

Microplastic contamination has emerged as a significant threat to freshwater ecosystems globally, yet data from African water bodies remain critically limited. This study assessed the occurrence of microplastics and their effects on water quality and oxidative stress responses in *Oreochromis niloticus* from Jakara Reservoir, Kano State, Nigeria. Physicochemical parameters including temperature, dissolved oxygen (DO), biochemical oxygen demand (BOD), total dissolved solids (TDS), electrical conductivity, turbidity, pH, phosphate-phosphorus, and nitrate-nitrites were analyzed across four sampling sites (A-D). Microplastic polymers (PET, PP, PVC, PA, PS) were identified in both surface water and fish tissues (gills and liver) using FTIR spectroscopy. Oxidative stress biomarkers—superoxide dismutase (SOD), catalase (CAT), and lipid peroxidation (LPO)—were measured in gill and liver tissues. Results revealed significant spatial variation in water quality parameters, with several parameters exceeding FAO/WHO guidelines, particularly TDS (733.8 ± 1.40 mg/L), electrical conductivity (1064 ± 0.16 μ S/cm), turbidity (38.9 ± 0.001 NTU), nitrate-nitrites (11.50 ± 0.74 mg/L), and phosphate-phosphorus (3.11 ± 0.01 mg/L). Microplastic abundance in surface water ranged from 10-45 particles per site, with PET and PP being most prevalent. Fish from Site D showed the highest microplastic accumulation (35 particles in gills, 13 in liver). Significant oxidative stress was observed, with maximum SOD (4.65 ± 0.78 U/mg prot.) and CAT (95.11 ± 0.01 U/mg prot.) activities in gills at Site C, while liver LPO peaked at Site D (2.36 ± 0.01 nmol/g). Correlation analysis revealed significant positive relationships between MPs and DO ($r=0.794$), turbidity ($r=0.789$), phosphate ($r=0.711$), and nitrate ($r=0.732$), with a strong negative correlation with TDS ($r=-0.839$). These findings demonstrate that microplastic pollution in Jakara Reservoir is associated with water quality degradation and oxidative stress in *O. niloticus*, posing risks to aquatic ecosystem health. It is therefore recommended that appropriate measures should be taken to curtail the menace that could disrupt this aquatic ecosystem in the long run.

Keywords: Microplastics, *Oreochromis niloticus*, Water quality, Oxidative stress, Jakara River, Nigeria

INTRODUCTION

Microplastic (MP) pollution has emerged as one of the most pressing environmental challenges of the 21st century. Defined as plastic particles smaller than 5 mm, microplastics originate from primary sources (manufactured microbeads, industrial pellets) and secondary sources (fragmentation of larger plastic debris). Global plastic production reached 359 million tonnes in 2018, with projections indicating that microplastic emissions into aquatic environments could reach 20-53 million tonnes annually by 2030. Freshwater ecosystems are particularly vulnerable to microplastic contamination, serving as both sinks and conduits for plastic pollution from terrestrial sources. Rivers transport an estimated 1.15-2.41 million tonnes of plastic waste annually from land to oceans, with African water bodies receiving substantial inputs from urban and industrial activities. Nigeria, as the ninth-largest plastic waste producer globally, generates approximately 2.5 million tonnes of plastic waste annually, with less than 12% recycled, leaving vast quantities uncollected or improperly managed (Ichado, et al., 2025).

The Jakara River, located in the densely populated Kano metropolitan area (over six million inhabitants), exemplifies the challenges facing African freshwater systems. The river serves as both a primary water source for the region and a major drainage channel for municipal sewage and industrial discharge. Previous studies have documented significant anthropogenic pollution in the Jakara River, with water quality parameters showing seasonal variation attributed to dry-season sewage inputs and wet-season runoff. *Oreochromis niloticus* (Nile tilapia) serves as an ideal sentinel species for ecotoxicological studies due to its widespread distribution, economic importance, and frequent use in aquaculture. As a primary protein source providing approximately 40% of animal protein intake for Nigerians, fish consumption patterns make microplastic contamination in tilapia a direct human health concern. Nile tilapia encounter microplastics through ingestion and gill ventilation, with particles translocating across biological barriers to various organs including gastrointestinal tract, liver, gills, and brain.

The toxicological mechanisms of microplastics in fish are multifaceted. Physical effects include gut obstruction, epithelial abrasion, and tissue damage, while biochemical effects involve oxidative stress, inflammation, endocrine disruption, and genotoxicity (Sani, et al., 2022). Microplastics induce oxidative stress by overwhelming endogenous antioxidant defenses (superoxide dismutase, catalase, glutathione peroxidase), leading to lipid peroxidation, protein oxidation, and DNA damage (Dinku, et al., 2025). These particles also act as vectors for adsorbed pollutants including heavy metals and persistent organic pollutants, creating complex toxic mixtures. In view of the forgoing, this study aimed at assessing and quantifying microplastic occurrence in water and *O. niloticus* of Jakara Reservoir, Kano. Nigeria

MATERIALS AND METHODS

Study Area

Jakara Reservoir is located in Kano State, northwestern Nigeria (latitude 12°10'N to 12°13'N, longitude 8°31'E to 8°45'E). The region experiences a sub-tropical climate with mean temperatures of 25°C and 28°C for winter and summer, respectively, and total annual rainfall of 800-1000 mm concentrated in the wet season (April-October). The reservoir was impounded in 1976 and the major tributaries of Jakara reservoir are river Jakara and river Getsi which owns significant ecological value (Magaji and Rabi, 2020). It received most of

Kano cities domestic and industrial waste water. The reservoir was constructed to provide for fish production and irrigation. The secondary benefit of the dam is for recreation and wide life conservation (WRECA, 1990). It was about 45km from state capital. Two distinct seasons are experience annually (wet and dry seasons) consisting of a long summer and short winter. The rainy season last from May to September and dry season runs from October to April (Badamasi, 2014).

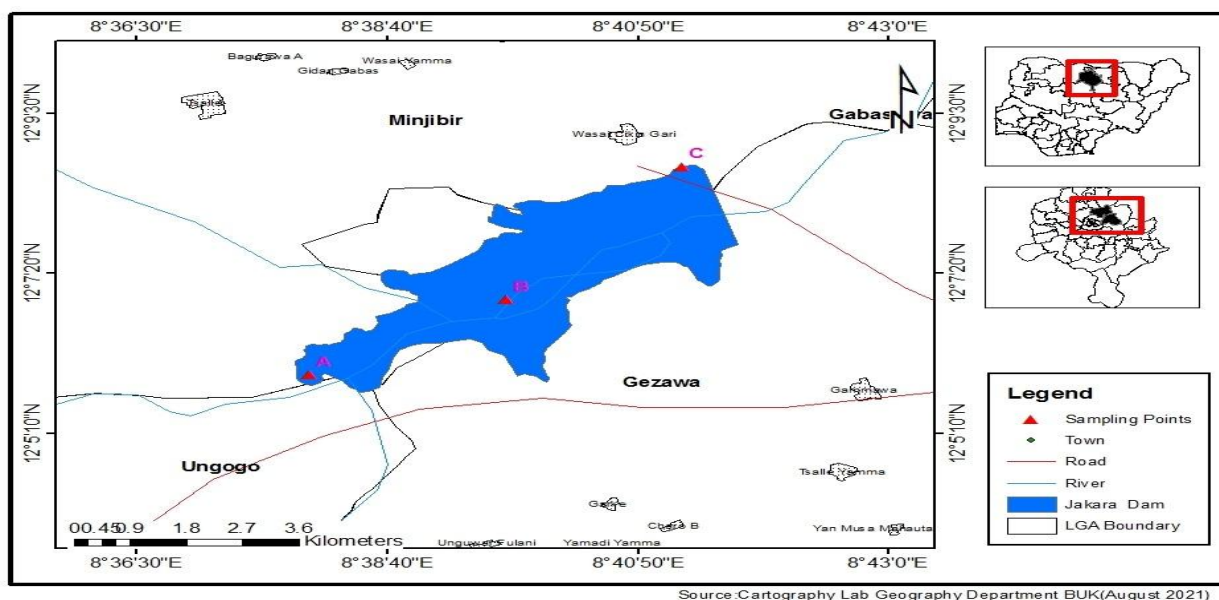


Fig 1: Study Area: Jakara Reservoir, Kano State, Nigeria indicating Sampling Sites (Source: Cartography Lab. BUK).

Sampling Design

Four sampling sites (A-D) were selected based on land use patterns and pollution sources. Water samples and *O. niloticus* specimens were collected using standard procedures. Physicochemical Parameters determination concentration of the following physico-chemical parameters were analyzed from the water: Temperature, pH, TDS, Turbidity, Electrical Conductivity, DO, BOD, Total Nitrate, phosphate-phosphorous using procedure described by FAO/WHO (2018).

Microplastic Analysis

Surface water samples (10-15 cm below surface) were collected in acid-washed glass containers. Fish specimens were dissected to extract gill and liver tissues. Two hundred and seventy (270) fish samples with mean weight of 343.20 ± 7.62 g and mean length of 32.03 ± 1.10 cm were examined for the presence of microplastics. They were weighted to the nearest 0.1g using weighing balance (M-Metlar Model). The Total Lengths were taken on measuring board from the tip of mouth to the distal tip of the longest caudal fin ray with the aid of calibrated meter rule to the nearest 0.1cm. Immediately after dissection, gastrointestinal contents from oesophagus to the vent, dorsal part and anus were transferred into 500 mL beaker and 200 mL of H_2O_2 (30% v/v) were added and shake vigorously. Assessment of microplastics content was carried out using gravimetric methods for each fish as described by Atamanalp, et al. (2021). Stomach contents were removed and incubated at 40 °C for 72hrs to digest the organic matter. To extract all types of microplastics, density

gradient separation techniques was carried out by the addition of 300mL of saturated sodium chloride solution (1.2 g/ mL NaCl) and kept for 8hrs at room temperature as described by Jabeen, *et al.* (2017). After incubation, a Polyester (PETE) membrane filter (10µm, 25mm diameter) was used to suspend the contents in an air-dried at room temperature for 4-5hrs.

Microplastics were isolated using density separation and filtration, followed by identification and characterization using Fourier-transform infrared spectroscopy (FTIR). MPs were identified based on their morphological features such as the types (fiber, film, fragment and beads) and the number of MPs particles per fish were counted as adopted by (Sun, *et al.*, 2019; Aytan, *et al.*, 2020). Chemical characterization (molecular bonds, chemical compounds and functional groups) of the identified MP particles were carried out using infrared radiation using Fourier Transform Infrared Spectrophotometer (The Cary 630 FTIR spectrometer Model) in the range 650-4000 cm⁻¹ at Biochemistry Laboratory, Bayero University, Kano using the procedure adopted by Mukund, *et al.* (2014).

Oxidative Stress Biomarkers

Gill and liver tissues were homogenized and centrifuged for biochemical analysis. Superoxide dismutase (SOD) and catalase (CAT) activities were measured spectrophotometrically as markers of antioxidant defense, while lipid peroxidation (LPO) levels were determined as an indicator of oxidative damage.

Statistical Analysis

Data were expressed as mean ± standard deviation. One-way ANOVA followed by post-hoc tests identified significant differences between sites ($p < 0.05$). Pearson correlation analysis determined relationships between physicochemical parameters and microplastic abundance. Canonical Correspondence Analysis (CCA) visualized associations between microplastics, sampling sites, and environmental variables.

RESULTS AND DISCUSSION

Table 1 presents the spatial variation of physicochemical parameters across sampling sites A to D, with comparative FAO/WHO (2018) guidelines. Water temperature values ranged from 22.41±0.14°C (Site D) to 28.3±0.41°C (Site B), all within the FAO/WHO (2018) guideline of <40°C. The lower temperature at Site D may reflect depth differences, groundwater influence, or shading effects. Significant differences between Sites B (28.3°C) and C (27.1°C) ($p < 0.05$) suggest localized thermal inputs from industrial discharges. Similar observation was reported by Adewunmi, *et al.* (2022). Dissolved Oxygen concentrations ranged from 5.23±0.10 mg/L (Site A) to 5.46±0.10 mg/L (Site D), all within the 5.0-9.0 mg/L of WHO (2023) guideline. The significantly higher DO at Sites C and D ($p < 0.05$) suggests greater oxygenation from mixing or photosynthetic activity, while lower DO at Site A might indicate organic matter decomposition or reduced aeration as reported by (Aina, *et al.*, 2020). Biochemical Oxygen Demand values ranged from 1.98±0.41 mg/L (Site D) to 3.7±1.00 mg/L (Site C). BOD exceeded the 3.0-6.0 mg/L guideline at Site C (3.7 mg/L), indicating significant organic pollution at this location. The high variability at Site C suggests pollution inputs as reported by Aytan, *et al.* (2020). Mustapha, *et al.* (2012) similarly found high BOD in the downstream Jakara Reservoir, attributing it to municipal sewage and industrial discharges.

Total Dissolved Solids ranged from 433.3 ± 0.10 mg/L (Site C) to 733.8 ± 1.40 mg/L (Site A), with all sites except C exceeding the FAO/WHO (2018) guideline of <500 mg/L. Site A showed the highest TDS, suggesting inputs from dissolved mineral salts, agricultural runoff, or sewage effluent. This is in-line with work of Ferreriera, *et al.* (2020). High TDS affects water palatability, osmotic regulation in aquatic organisms, and indicates contamination from anthropogenic sources. Electrical conductivity ranged from 637.1 ± 0.14 μ S/cm (Site C) to 1064 ± 0.16 μ S/cm (Site A), with Sites A and D exceeding the <1000 μ S/cm guideline. The strong correlation between EC and TDS ($r=0.459$) reflects dissolved ionic content. High EC at Sites A and D indicates substantial dissolved salts, potentially from sewage, agricultural runoff, or industrial effluent. Mustapha, *et al.* (2012) found EC as a key parameter distinguishing pollution sources in Jakara River.

Turbidity ranged from 26.9 ± 0.03 NTU (Site C) to 38.9 ± 0.001 NTU (Site A), all exceeding the <25 NTU WHO (2023) guideline. Elevated turbidity indicates suspended particulate matter, which reduces light penetration, affects primary productivity, and can carry adsorbed pollutants including microplastics. The significantly lower turbidity at Site C (26.9 NTU) may reflect sedimentation processes or reduced inputs at this location (Qiu, 2015). pH ranged from 7.8 ± 0.11 (Site A) to 8.6 ± 0.52 (Site D), all within the 6.0-9.0 WHO (2023) guideline. The slightly alkaline pH at all sites indicates buffering capacity from carbonate or sewage inputs. Site D showed the highest pH (8.6) and greatest variability (± 0.52), indicating pollution influence. This is in-line with work of Guven, *et al.* (2022).

Phosphate-Phosphorus ranged from 1.87 ± 0.54 mg/L (Site D) to 3.11 ± 0.01 mg/L (Site B), all far exceeding the 0.1 mg/L FAO/WHO (2018) guideline. High phosphate concentration indicates severe nutrient pollution, primarily from sewage, agricultural fertilizers, and detergent inputs as reported by Bour, *et al.* (2018). Phosphate enrichment promotes eutrophication, leading to algal blooms, oxygen depletion, and ecosystem degradation. The significant positive correlation between phosphate and microplastics ($r=0.711$) suggests co-transport or shared pollution sources. This is in-line with the work Guven, *et al.* (2022). Nitrate-Nitrites ranged from 5.26 ± 0.10 mg/L (Site C) to 11.50 ± 0.74 mg/L (Site D), with Sites A, B, and D exceeding the 5 mg/L FAO/WHO (2018) guideline. High nitrate indicates nitrogen pollution from sewage, agricultural runoff, and industrial effluent. The highest concentration at Site D (11.50 mg/L) suggests significant nitrogen loading at this location, with implications for eutrophication and human health (methemoglobinemia risk). The significant positive correlation between nitrate and microplastics ($r=0.732^*$) indicates shared anthropogenic sources as reported by Merga, *et al.* (2020) and Aytan, *et al.* (2020).

Table 1: Spatial Variation of Physicochemical Parameters obtained from Jakara Reservoir, Kano State

Sampling Sites	A	B	C	D	FAO/WHO (2018)
Water temperature (°C)	27.8±±0.16 ^a	28.3±0.41 ^a	27.1±0.04 ^b	22.41±±0.14 ^a	<40°C mg/L
DO (mg/L)	5.23±0.10 ^a	5.39±0.12 ^a	5.45±0.16 ^b	5.46±0.10 ^a	5.0-9.0 mg/L
BOD (mg/L)	2.4±0.21 ^a	2.9±0.01 ^b	3.7±1.00 ^a	1.98±0.41 ^b	3.0-6.0 mg/L
TDS (mg/L)	733.8±1.40 ^a	529.1±0.66 ^b	433.3±0.10 ^a	654.3±5.76 ^b	<500 mg/L
Electrical Conductivity (µS/cm)	1064±0.16 ^a	875.3±1.61 ^b	637.1±0.14 ^a	987.4±7.05 ^{ab}	<1000 µ/Scm
Turbidity (NTU)	38.9±0.001 ^a	31.2±0.11 ^a	26.9±0.03 ^b	35.7±0.75 ^a	<25 NTU
pH	7.8±0.11 ^a	7.9±0.10 ^b	8.1±0.01 ^a	8.6±0.52 ^a	6.0-9.0
Phosphate - phosphorus (mg/L)	2.12±0.10 ^b	3.11±0.01 ^a	2.28±0.21 ^b	1.87±0.54 ^b	0.1 mg/L
Nitrate - nitrites (mg/L)	6.11±0.53 ^a	7.01±0.81 ^a	5.26±0.10 ^b	11.50±0.74 ^a	5 mg/L

Values are means and S.D, values with the different superscript in a column differed significantly (p<0.05)

Table 2 revealed microplastic contamination in both gills and liver of *O. niloticus* across all sampling sites. Microplastic abundance per site ranged from 7 (Site A, liver) to 35 (Site D, gills). Gill tissue showed higher microplastic abundance than liver tissue across all sites. This pattern might reflect gills' direct and continuous contact with water during respiration, making them a primary entry point for microplastics. The gill's large surface area and high filtration rate facilitate particle capture and retention (Baalkhuyur, et al., 2018). Site D showed the highest microplastic concentration (35 particles in gills, 13 in liver), which is in consistent with its high nitrate (11.50 mg/L) and pH (8.6) levels, suggesting shared pollution sources. This is in agreement with the works of Ajegi, et al. (2025); Ichado, et al. (2026). The detection of microplastics in both surface water and fish tissues confirms that the reservoir is contaminated and that biota are exposed to and accumulating plastic particles. The progressive increase in microplastic abundance from Sites A to D indicates spatial accumulation of pollution along the Jakara reservoir as reported by Yahaya, et al. (2026) in Sokoto State.

Table 2: Microplastics Occurrence In Different Body Parts of *Oreochromis niloticus* obtained from Jakara Reservoir

Sam pling site	Part of fish collected	Size (mm)	Mp type	No. of MPs/organ	Source
A	Liver	0.16±0.01	PET, PS, PP	7	Soft drinks, water bottle, plastic containers
	Gill	0.14±0.00	PVC	21	Abrasion from car tires
B	Gill	0.28±0.01	PVC	18	Electric cables, Abrasion from car tires
	Liver	0.18±0.00	PA,	8	Thread, soft drink
C	Gill	0.44±0.10	PA, PET	26	plastic containers
	Liver	0.39±0.00	PET, PE	11	Water bottle
D	Gill	0.34±0.10	PS	35	plastic containers, water bottle,
	Liver	0.32±0.00	PP	13	Electric cables,

Polystyrene (PS), Polyethylene (PE), Polyethylene terephthalate (PET), polyvinyl chloride (PVC), Polypropylene (PP)

Six polymer types were identified across sampling sites with PET, PS, PP (Site A); PVC (A, B); PA, PET (B, C); PE (C); PP (D). PET and PP had highest abundance, consistent with other African freshwater microplastic studies Dzierżyński, et al. (2025). These polymers dominate due to their extensive use in beverage bottles, food containers, and packaging materials. PVC detection at Sites A and B suggests inputs from electric cable abrasion and construction materials. This is in agreement with the work of Ngomane, et al. (2026). PA (nylon) at Sites B and C indicates contributions from synthetic textiles and fishing gear. Microplastic sizes ranged from 0.14±0.00 mm (Site A, gills) to 0.44±0.10 mm (Site C, gills). Gill particles generally showed larger sizes (0.14-0.44 mm) compared to liver particles (0.16-0.39 mm), suggesting that larger particles are trapped in gills while smaller particles (≤0.39 mm) can translocate across biological barriers to reach the liver. This observation is in consistent with the finding of Sani, et al. (2022) and Pradit, et al. (2023). The smallest particles (0.14-0.18 mm) were PS and PET, indicating that polymer type influences fragmentation patterns and bioavailability. Identified sources of microplastics included soft drinks and water bottles (PET), plastic containers (PET, PP), abrasion from car tires (PVC), electric cables (PVC, PP), and synthetic threads (PA). This diverse source profile reflects the multiple anthropogenic inputs to Jakara Reservoir, which is in consistent with the river's role as a drainage channel for the Kano metropolitan area. Similar observation was reported by Abdullahi, et al. (2025) in the same reservoir. The dominance of PET and PP aligns with findings from African freshwater systems where these polymers are frequently dominant (Atukwatse, et al., 2026). Dada and Bello (2023) similarly found PET and PP in Lagos

Lagoon, though fibers dominated water samples while fragments dominated sediments. Spatial variation revealed that site D had the highest abundance for multiple polymer types indicating it as a pollution hotspot. Microplastic concentrations in surface water indicate that the reservoir serves as both a sink and conduit for plastic pollution. The presence of multiple polymer types reflects diverse sources including packaging materials, textiles, construction materials, and vehicle wear. These findings raise concerns about ecosystem health and human exposure through water use and fish consumption.

Table 3: Mean Abundance of microplastics in surface water samples of Jakara Reservoir, Kano, Nigeria

Microplastics polymers	Site A	Site B	Site C	Site D	Site E	Site F
PET	15±0.17	19±1.09	34±1.08	45±0.65	23±2.98	33±1.98
PP	12±0.50	23±0.65	34±0.34	19±0.54	34±1.98	34±0.98
PVC	11±0.22	23±0.98	25±0.19	34±1.08	22±0.67	32±0.65
PA	10±0.50	18±0.54	19±0.88	34±0.46	29±1.65	21±0.21
PS	10±0.10	20±0.32	13±0.45	28±1.87	29±2.98	34±0.65

Note: polypropylene (PP), polyethylene (PE), polyvinyl chloride (PVC), polyethylene terephthalate (PET) and polystyrene (PS)

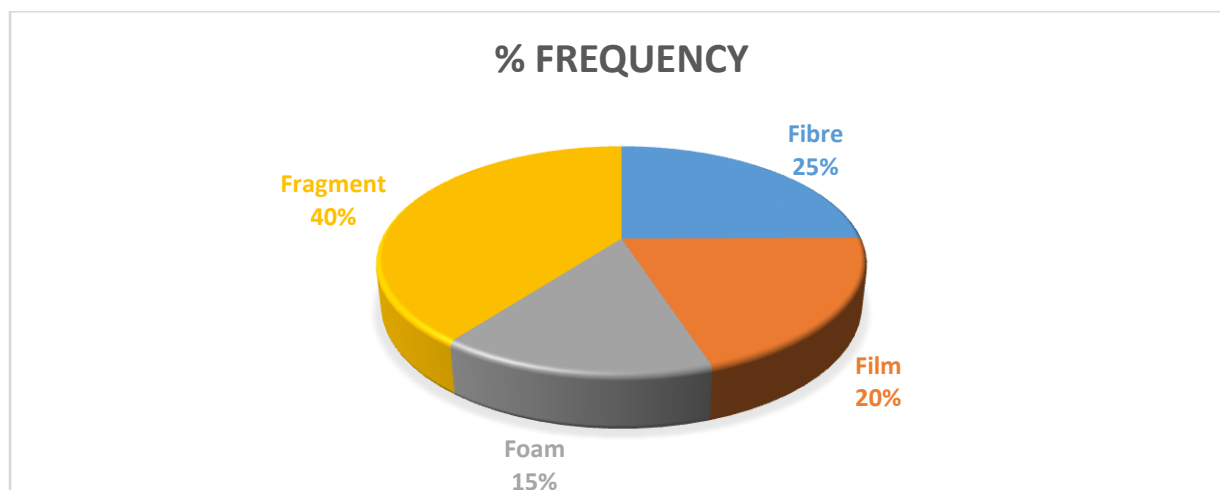


Figure 2: percentage frequency of different microplastics shapes examined in surface water samples from Jakara Reservoir

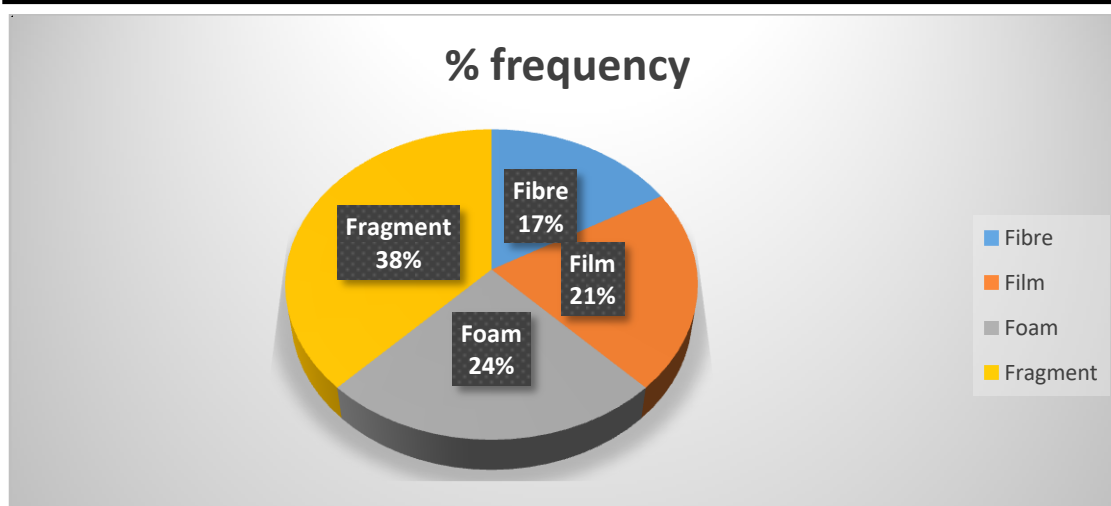


Figure 3: Percentage Frequency of Microplastics Shapes Recovered from Fish samples

Superoxide dismutase concentrations in the gill samples indicate that SOD ranged from 2.31 ± 0.01 U/mg prot. (Site A) to 4.65 ± 0.78 U/mg prot. (Site C), showing significant spatial variation ($p < 0.05$). Liver SOD ranged from 1.66 ± 0.41 U/mg prot. (Site A) to 2.74 ± 0.01 U/mg prot. (Site C). Site C showed the highest SOD activity in both gills and liver, suggesting a high oxidative stress at this location. The significantly high SOD at Site C coincides with high BOD (3.7 mg/L), indicating that organic pollution exacerbates oxidative stress. The high SOD activity indicates increased superoxide radical production, triggering up regulation of antioxidant defense mechanisms. This is in-line with the finding of Usman, et al. (2026). Studies on Nile tilapia exposed to polystyrene microplastics showed significant SOD up regulation (2.55-fold to 3.80-fold) in blood cells, confirming that MPs induce oxidative stress through reactive oxygen species generation. Similarly, combined exposure to polystyrene nanoplastics caused significant oxidative stress in the sampled fish's liver and gills. CAT works in concert with SOD to detoxify hydrogen peroxide produced from superoxide dismutation. The high CAT activity at Site C suggests increased H_2O_2 production, from microplastic-induced oxidative stress. Wangboje, et al. (2024) found that microplastics cause oxidative stress in African freshwater fish (*O. niloticus*) through similar mechanisms. CAT concentration in the gill tissues ranged from 78.02 ± 1.50 U/mg prot. (Site A) to 95.11 ± 0.01 U/mg prot. (Site C), with all sites showing high activity. Liver CAT ranged from 71.90 ± 0.10 U/mg prot. (Site A) to 89.48 ± 0.03 U/mg prot. (Site C). Site C had the highest CAT activity, which is consistent with the SOD pattern and confirming Site C as a high-stress location.

LPO concentration ranged from 1.75 ± 0.01 nmol/g (Site A) to 3.47 ± 0.21 nmol/g (Site B). Liver LPO ranged from 0.21 ± 1.31 nmol/g (Site A) to 2.36 ± 0.01 nmol/g (Site D). Gill tissue had a higher LPO than liver, indicating greater oxidative damage in gills. Site B showed the highest gill LPO (3.47 nmol/g), while Site D showed the highest liver LPO (2.36 nmol/g), suggesting organ-specific damage. LPO measures oxidative damage to cellular membranes through polyunsaturated fatty acid peroxidation. High LPO indicates that antioxidant defenses (SOD, CAT) are insufficient to prevent oxidative damage. Studies by Merga, et al. (2020) reported positive associations between microplastic burden and oxidative stress biomarkers, with spatial and seasonal variation. Similarly, Nile tilapia studies have documented increased LPO following microplastic exposure. Gills showed higher SOD, CAT, and LPO than liver across all sites, indicating gills are more vulnerable to oxidative stress. This vulnerability reflects the gill's direct contact with waterborne pollutants, high blood flow, and large surface area, which facilitate pollutant uptake and oxidative damage.

This is in-line with findings of Ajegi, et al. (2025); Ichado, et al. (2026) and Usman, et al. (2026).

Table 4: Oxidative Stress Enzymes Activity of Antioxidant Enzymes in *O. niloticus* from Jakara Reservoir, Kano State

Organ	Sites/Activity	SOD (unit/ mg prot.)	CAT (Unit/mg prot.)	LPO(nmol/g)
Gills	A	2.31 ±0.01 ^b	78.02±1.50 ^a	1.75 ±0.01 ^b
	B	3.41±0.22 ^b	92.80±0.10 ^a	3.47±0.21 ^a
	C	4.65±0.78 ^a	95.11±0.01 ^a	2.87±0.01 ^a
	D	2.64±0.07 ^a	87.56±1.16 ^a	2.76±0.11 ^a
Liver	A	1.66 ±0.41 ^b	71.90±0.10 ^a	0.21 ±1.31 ^a
	B	2.27±0.17 ^a	84.32±0.05 ^a	1.86±1.03 ^a
	C	2.74±0.01 ^a	89.48±0.03 ^a	2.20±0.01 ^a
	D	1.68±0.12 ^b	72.5±1.67 ^a	2.36±0.01 ^a

Mean values with different superscripts in a column are significantly different (P<0.05)

Besides, with regards to correlation matrix, microplastics showed significant positive correlations with DO (r=0.794), turbidity (r=0.789), phosphate (r=0.711), and nitrate (r=0.732), while showing a strong negative correlation with TDS (r=-0.839) (Table 5). These correlations indicate that microplastic abundance is associated with water quality parameters indicative of pollution as reported by Mendoza-López, et al. (2025). The positive correlation between MPs and turbidity (r=0.789) is consistent, as suspended particles (including microplastics) contribute to turbidity. Microplastics adsorb to suspended sediments, creating aggregates that increase light scattering (Zeenat, et al., 2021). The positive correlations with phosphate and nitrate confirm that microplastics co-occur with nutrient pollution, indicating shared anthropogenic sources (sewage, agricultural runoff, industrial effluent). The strong negative correlation between MPs and TDS (r=-0.839) is notable, as both are indicators of anthropogenic pollution.

Table 5: Pearson Correlation Coefficient Showing Relationship between Physico-chemical parameters and Microplastics Obtained from Jakara Reservoir, Kano

Parameters	Temp.	pH	DO	BOD	TDS	EC	Turbidity	PO ₃ ⁻⁴	N03	MPs
Temp.	1.000									
Ph	0.760*	1.000								
DO	0.723*	0.317	1.000							
BOD	-0.343	-0.451	0.745*	1.000						
TDS	0.567	-0.145	0.343	0.748	1.000					
EC	0.565*	0.671*	0.356	0.751*	0.459	1.000				
Turbidity	0.331	0.432	0.652*	0.343	-0.217	0.651*	1.000			
PO ₃ ⁻⁴	0.534	0.743*	0.621*	0.432	0.801*	0.342	-0.202	1.000		
No3	0.673*	0.834*	0.563*	0.215	0.398	0.354	0.214	0.230	1.000	
MPs	0.321	0.130	0.794*	0.321	-0.839	0.240	0.789*	0.711*	0.732*	1.000

*Correlation Significant at 0.05 (2-tailed)

** Correlation Significant at 0.01 (2- tailed)

CCA biplot indicates associations between microplastics, sampling sites, and physicochemical parameters (Fig. 4). The CCA ordination shows Sites A and B clustering together, separated from Sites C and D, indicating spatial gradients in pollution characteristics as observed by Dinku, et al. (2025). The CCA results support the Pearson correlation findings and suggest that microplastic distribution is controlled by multiple environmental factors including hydrology, pollution sources, and sediment characteristics. Sites C and D show a stronger association with microplastics, turbidity, DO, BOD, phosphate, and nitrate, indicating multiple pollution sources at these locations. Site A may show stronger association with TDS and EC, suggesting different pollution mechanisms (dissolved salts vs. particulate plastics). This spatial separation confirms that microplastic contamination varies significantly with location and associated pollution sources. This is in agreement with the work of Sani, et al. (2022).

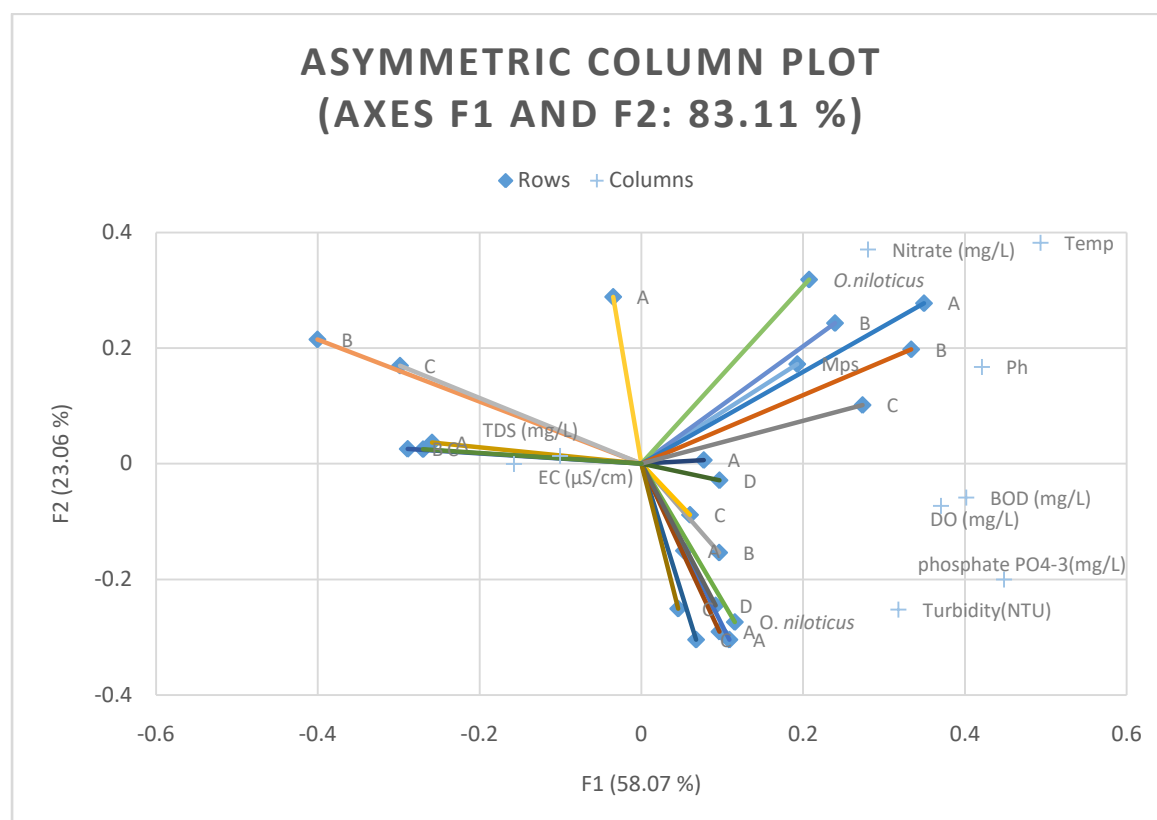


Fig. 4: Canonical Correspondence Analysis (CCA) showing association between Microplastics contents, sampling sites and Physicochemical Parameters in Jakara Reservoir, Kano State

CONCLUSION

This study provides comprehensive data on microplastic contamination in Jakara Reservoir and its association with water quality degradation and oxidative stress in *Oreochromis niloticus*. Physicochemical parameters, particularly TDS, EC, turbidity, phosphate, and nitrate, exceed FAO/WHO (2018) guidelines, indicating severe anthropogenic pollution from municipal sewage, agricultural runoff, and industrial effluent. The Jakara River receives substantial pollutant inputs from the densely populated Kano metropolitan area. Microplastics (PET, PP, PVC, PA, PS) are present in both surface water and fish tissues, with significant spatial variation. Site D is a pollution hotspot with the highest microplastic abundance and nutrient concentrations. Significant oxidative stress in *O. niloticus* shows high antioxidant

enzyme activities and lipid peroxidation indicative of oxidative stress. Gills show greater vulnerability than liver, and responses differ spatially with pollution levels. Positive correlations between microplastics and, TDS, turbidity, phosphate, and nitrate indicate shared anthropogenic sources, while the negative correlation with DO suggests different pollution mechanisms.

RECOMMENDATIONS

1. Further studies on other pollutants such as hydrocarbons, heavy metals and algal blooms should be carried out.
2. Toxicological assessment of microplastics on humans that consumed the water and the fishes from River Wudil should be carried.
3. Improve municipal and industrial wastewater treatment infrastructure to remove microplastics and nutrients through advanced technologies such as membrane filtration, coagulation-flocculation) should be adopted.

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